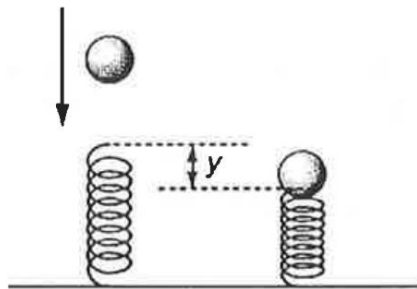


- 7 A ball of mass m falls freely from rest as shown below. When it has reached a speed v , it just strikes a vertical spring. The spring is then compressed by a distance y before the ball moves upwards again.



Assumes that all the energy the ball loses becomes elastic potential energy in the spring.

What is the average force exerted by the spring during its compression.

- A $\frac{mv^2}{2y}$ B $\frac{m}{2y}(v^2 - 2gy)$ C $\frac{mv^2}{y}$ D $\frac{m}{2y}(v^2 + 2gy)$

Ans: D

$$\frac{1}{2}Fy = \frac{1}{2}mv^2 + mgy$$

$$F_{\text{ave}} = \frac{1}{2}F = \frac{\frac{1}{2}mv^2 + mgy}{y}$$

- 8 The driver of a 1000 kg car tries to turn through a circle of radius 100 m on an unbanked curve